

1. Introduction

AgriCycle is an Android mobile application that connects suppliers of agricultural waste with buyers who are interested in purchasing agricultural by-products. The app is built using Android Studio and Firebase (Authentication and Realtime Database). This manual explains how to install the app, sign in, and use the three main features implemented in Phase 3:

- Supplier adds a waste listing.
- Buyer browses listings and makes an offer or buys directly.
- Supplier manages offers and views accepted deals.

2. Getting Started

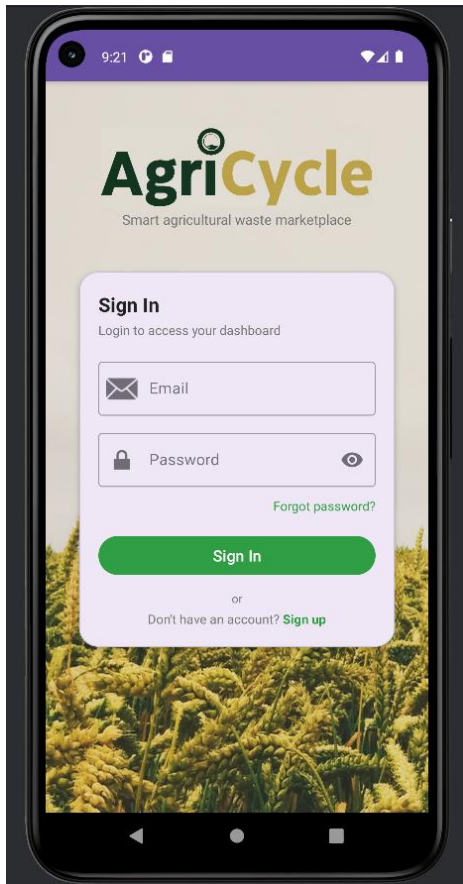
2.1 Installation

1. Copy the AgriCycle APK file to your Android device or run the project from Android Studio.
2. On the device, open the APK and allow installation from unknown sources if prompted.
3. After installation, the AgriCycle icon appears in the app list.

2.2 Launching the App and Signing In

1. Tap the AgriCycle icon to launch the app.
2. The Sign In screen appears. Enter your registered email and password.
3. Tap “Sign In” to access your dashboard. If you do not have an account, tap “Sign up” to create one.
4. If you forget your password, use the “Forgot password?” link to reset it.

Figure 1 – Sign In screen

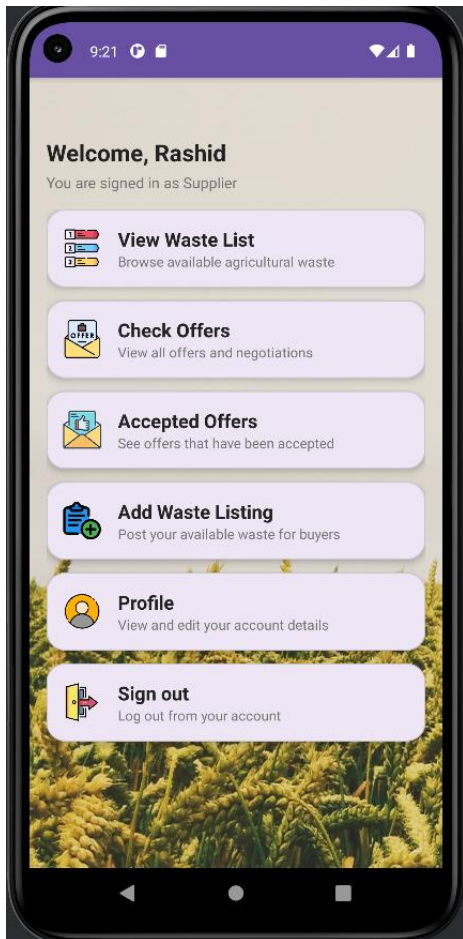


3. Dashboard Overview

After a successful login, users see a dashboard. The options depend on the role. In Phase 3 we focus on the Supplier role, which has the following options:

- View Waste List – browse all waste listings in the system.
- Check Offers – view pending offers on your listings.
- Accepted Offers – view offers that you have accepted.
- Add Waste Listing – create a new listing as a supplier.
- Profile – view your profile data and rating.
- Sign out – log out from the application.

Figure 2 – Supplier dashboard



4. Use Case 1 – Supplier: Add Waste Listing

4.1 Purpose

Allows a supplier to publish a new agricultural waste item with a name, type and price.

4.2 Preconditions

- The user is logged in as a Supplier.
- The device has an active internet connection.

4.3 Steps

1. From the dashboard, tap “Add Waste Listing”.
2. The Add Waste Listing screen appears.
3. Verify the Supplier field (your name) is correct.
4. Enter the Waste name (for example, “Palm leaves”).
5. Enter the Type (for example, “organic”, “biomass”, “dry”).
6. Enter the Price (AED) as a number.
7. Tap “Save listing”.
8. If the data is valid, a success message appears and the listing is stored in Firebase.

Figure 3 – Add Waste Listing screen

The screenshot shows a mobile application interface for adding a waste listing. At the top, the status bar displays the time 9:22 and various icons. The app's title bar is purple. Below the title bar, the screen is titled "Add Waste Listing". A light purple form is centered on the screen, containing the following elements:

- A label "Supplier" followed by the text "Rashid".
- A text input field labeled "Waste name".
- A text input field labeled "Type (e.g. organic, biomass, dry)".
- A text input field labeled "Price (AED)".
- A green button labeled "Save listing".

The background of the app is a field of yellow flowers. At the bottom of the screen, there are three navigation icons: a back arrow, a home circle, and a recent apps square.

4.4 Result

The new listing is stored in the database with status “available” and becomes visible in the Waste Listings screen for buyers and suppliers.

5. Use Case 2 – Buyer: Browse Listings and Make Offer / Buy Direct

5.1 Browse listings

1. From the dashboard, tap “View Waste List”.
2. The Waste Listings screen appears, showing all available items.

Figure 4 – Waste Listings screen



5.2 Make an offer

1. Choose a listing and enter your proposed price in the offer field (if available).
2. Tap “Make Offer”.
3. If the field is empty, the app displays “Enter amount first”.
4. If the amount is valid, the offer is submitted and the listing status becomes “pending”.

5.3 Buy directly

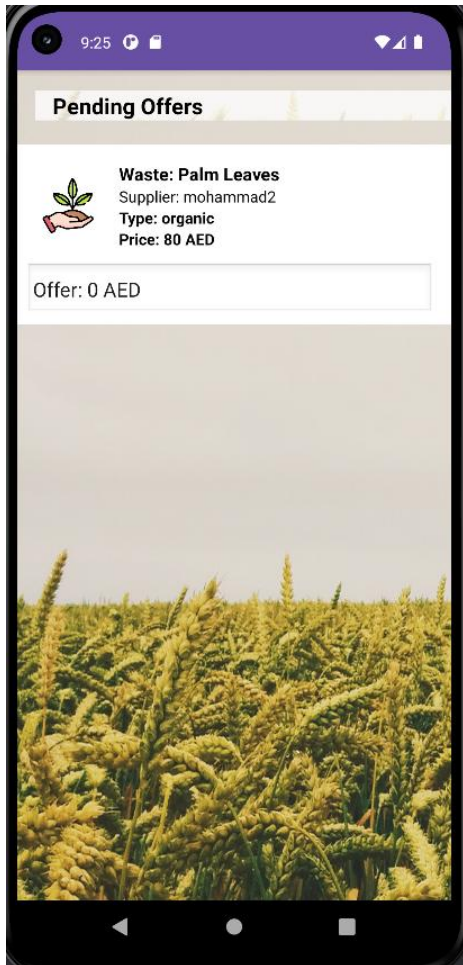
1. Tap the “Buy directly” button on the desired listing.
2. Confirm the purchase in the dialog.
3. The listing status becomes “sold”, the buyer is stored, and the deal value equals the listed price.

6. Use Case 3 – Supplier: Manage Offers and Accepted Deals

6.1 View pending offers

1. From the dashboard, tap “Check Offers”.
2. The Pending Offers screen appears and shows offers buyers have submitted on your listings.

Figure 5 – Pending Offers screen



3. For each offer you can:

- Accept – the listing status changes to “sold” and the deal is confirmed.
- Reject – the listing returns to “available” and the offer is cleared.

6.2 View accepted offers

1. From the dashboard, tap “Accepted Offers”.
2. The Accepted Offers screen shows all completed deals with the final agreed price.

Figure 6 – Accepted Offers screen (multiple deals)

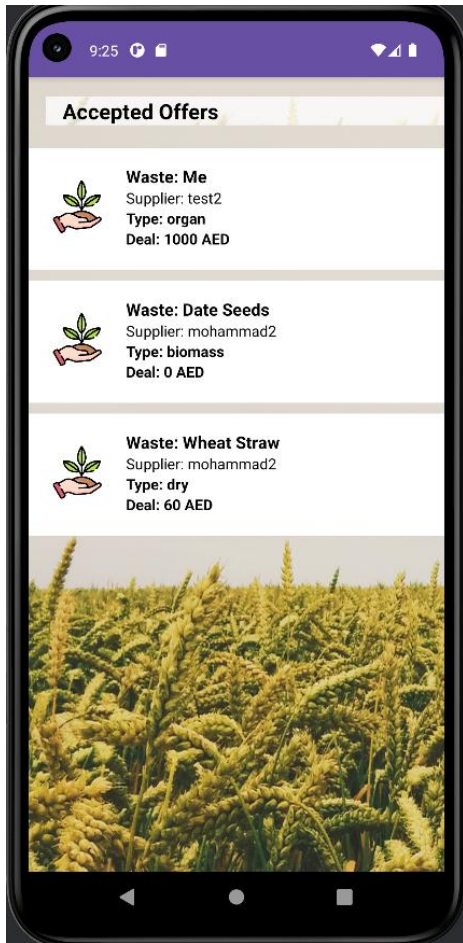
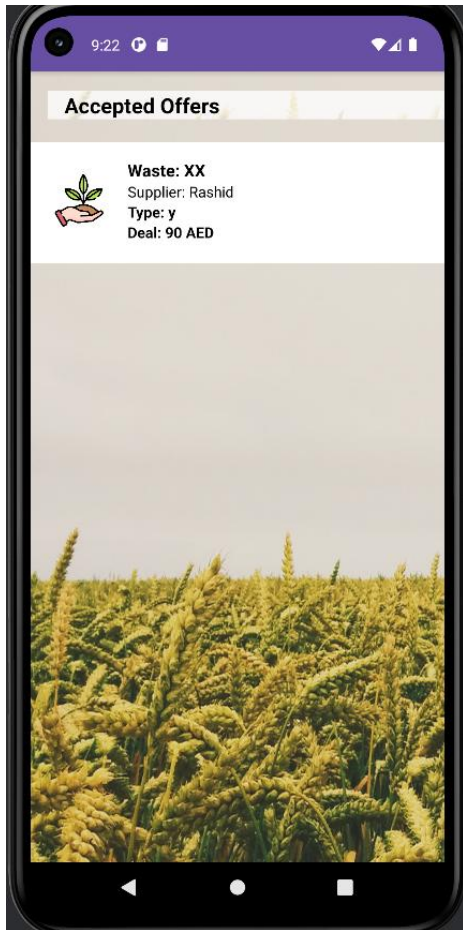


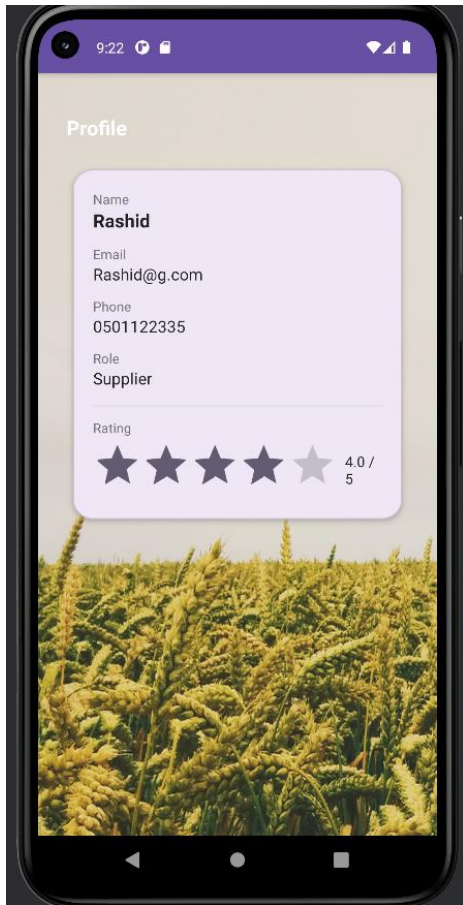
Figure 7 – Accepted Offers screen (single deal example)



6.3 Profile and rating

The Profile screen shows your name, email, phone number, role and average rating. Buyers may navigate to a supplier's profile from accepted deals to view or give feedback.

Figure 8 – Profile screen



7. Troubleshooting

- If you cannot sign in, check your email and password and verify you have an internet connection. Use “Forgot password?” if needed.
- If a listing or offer does not appear, refresh the screen or reopen the app.
- If the app crashes, restart it. If the problem continues, report the steps to the development team.